Nicholas Wanner

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SUMMARY

Mid-level software engineer with 3 years of R&D experience across automation, firmware, networking, microservices, and AI/ML, with 5 patents awarded for innovative solutions. Proven ability to lead projects at Dell, eager to drive cloud-focused initiatives.

EDUCATION

Bachelor of Science in Computer Engineering

Minor in Cyber Security • Texas A&M University • College Station, Texas • 2021 • 3.95

EXPERIENCE

Software Engineer

Dell Technologies - Innovation Studio — **June 2022 - September 2023, Austin, TX**

- Pioneered the design and adoption of Databricks MLOps stacks for pipeline templatization, configured to be config-driven for seamless adoption by data scientists; migrated 5 working models to the new system, streamlining deployment and cutting setup time by 30%.
- Spearheaded the development of a config-driven model optimization service utilizing ONNX Runtime and Hugging Face; boosted inference speeds by 15% and reduced storage requirements by 50% on client devices.
- Developed the edge component of a model management framework, setting up an edge server with REST API endpoints to offload client inference to the edge; optimized models using ONNX Runtime and other optimization tools, enhancing inference speed and reducing client device workload.
- Architected and implemented communication and state machine logic for a Wi-Fi-based Dell gaming controller, enhancing signal management; contributed to a CES award-winning solution and secured 2 patents for network resource distribution.
- Designed and developed an algorithm using a knapsack-like approach for a peripheral-to-system mapping service, optimizing port-to-system allocations based on user setups; built a microservice-based API consumption layer to support unboxing workflows and cart checkouts, enhancing user understanding of workstation setups and reducing item returns due to incompatibility by a predicted 10%.

Software Engineer

Dell Technologies - Technology Strategy Team — **June 2021 - June 2022, Austin, TX**

- Built a configuration-driven automation framework for AI model data collection, automating workload orchestration, settings adjustments, and telemetry capture; enabled 24/7 data collection, increasing speed by 3x and reducing future workload automation time by 75%.
- Developed a proof of concept for a proprietary Dell Bluetooth pairing experience, enhancing user interaction by automating device recognition; enabled pre-pairing at checkout and created a secure, simplified pairing process for Dell on Dell devices, cutting user interaction time and eliminating user-driven actions required for the pairing process by 90%.
- Developed a proof of concept for interference avoidance in high-density ad-hoc mesh networks for wireless docks, exploring centralized, decentralized, and hybrid configurations; enabled docks to self-organize and optimize bandwidth, reducing interference in crowded environments, resulting in 2 patents.
- Engineered an experimental solution to minimize packet jitter for TCP/UDP transmissions during network-intensive workloads, achieving a 95% reduction in jitter with minor latency trade-offs.

Software Engineering Intern

Dell Technologies - Technology Strategy Team — **June 2020 - August 2020, Austin, TX**

- Led the development of a proof of concept for a Collaboration Touchpad with integrated conferencing controls, collaborating with mechanical, electrical, firmware, and UX teams to design HID functionalities; facilitated integration with Zoom, Microsoft Teams, and other conferencing software, earning a patent and accelerating time to market by 50% with deployment in customer devices within a year.

Software Engineer Intern

ThoughtTrace — **June 2018 - August 2018, Houston, TX**

- Developed a web app for real-time data labeling and QA, boosting efficiency by 35% and replacing manual CSV batch processes with direct integration into the training data pool.
- Hand-labeled training samples for reinforcement learning, enhancing model robustness and increasing accuracy by 8% through precise annotation of complex lease documents.

SKILLS

Languages: Python, C/C++, C#, JavaScript, SQL
Frameworks & Tools: ASP.NET, Vue.js, Express.js, React, GraphQL, Docker, Kubernetes, Git, Jenkins, Azure, WSL, SQLAlchemy, Alembic, RESTful APIs, pytest, Databricks, Spark, TensorFlow, Hugging Face, ONNX Runtime, Airflow, MLflow
Techniques & Concepts: Agile/Scrum, Object-Oriented Programming (OOP), Version Control, CI/CD, API Design, System Design, Cross-Functional Collaboration, Code Review, Security, ETL, Data Processing